



SOP #19

Image Lab Software

Data processing & Quantitative Analysis using Image Lab Software from Bio-Rad:

1. Export the VersaDoc image to ImageLab (available for free download onto personal computers):
 - a. Save the file in .1sc format as the name of the imaged protein followed by a space and the date in format YYYY.MM.DD hh.mm, for the time of imaging. (PsbA 2014.08.20 14.20, for example)
 - b. Also save a .jpg file of the image with the same name.
2. Using “volume tools”, draw rectangles around all relevant bands, as well as one large rectangle around a relatively homogenous part of background. The “disc” parameter should be set to “2” (under the “lanes” tab)
3. Double-clicking on the rectangles, assign the “Background” rectangle and “Unknown” rectangles.
4. Change the background subtraction to global (radio button on left), and the units for standards to fmol.
5. Identify your standards under the “standards” radio button. Label all 3 to 5 rectangles with their appropriate fmol values (ex: 1500 fmol, 750 fmol, 375 fmol, 187.5 fmol for RbcL serial dilution).
6. Export the analysis table to Excel. Create a new folder in Dropbox and name it with the date in format YYYY.MM.DD. In this folder, save the Image Lab Excel output in .scn format. See SOP#24 for appropriate curve fits for further data analysis, if linear regression doesn't work well.
7. Save .jpg and .1sc files originally exported from the VersaDoc in the same folder as the .scn file.